```
-- Step 1 : Create database (optional, for organization)
CREATE DATABASE IF NOT EXISTS sagar_ass9;
USE sagar_ass9;

DROP TABLE IF EXISTS areas;

CREATE TABLE areas(
   radius DECIMAL(5,2),     -- Radius value (e.g., 5.00, 6.00, etc.)
   area   DECIMAL(10,2)     -- Computed area of circle
);

DELIMITER //

CREATE PROCEDURE calculate_area()
BEGIN
   -- Declare local variables
   DECLARE radius DECIMAL(5,2);
   DECLARE area DECIMAL(10,2);
   DECLARE i INT DEFAULT 5;        -- Starting radius value

   -- WHILE loop runs from radius = 5 to 9
   WHILE i <= 9 DO
      SET radius = i;                        -- Assign radius
      SET area = 3.14159 * radius * radius;      -- Area = πr²
      INSERT INTO areas(radius, area)           -- Store result
      VALUES (radius, area);
      SET i = i + 1;                         -- Increment radius
   END WHILE;
END //
DELIMITER ;

CALL calculate_area();

SELECT * FROM areas;
```

-- ----------
-- 💬 Viva Preparation Notes
-- ----------
-- 🔹 1️⃣ What is a Stored Procedure?
-- ➤ A stored block of SQL statements that can be executed by a single call.

-- 🔹 2️⃣ What does WHILE do in MySQL?
-- ➤ WHILE repeatedly executes statements as long as a condition is TRUE.

-- 🔹 3️⃣ Why use DECLARE inside BEGIN...END?
-- ➤ Variables must be declared inside the procedure block.

-- 🔹 4️⃣ Formula used for area:
-- ➤ Area = $\pi \times r^2$ ($\pi \approx 3.14159$)

-- 🔹 5️⃣ What is the purpose of the INSERT statement here?
-- ➤ To store each radius and its corresponding calculated area into the table.

-- 🔹 6️⃣ What happens if you call the procedure again?
-- ➤ It inserts the same set of radius–area pairs again (duplicates).

-- 🔹 7️⃣ How to avoid duplicates?
-- ➤ You can TRUNCATE the table before calling the procedure again.

-- 🔹 8️⃣ Example Output:
```
-- +--------+--------+
-- | radius | area   |
-- +--------+--------+
-- |   5.00 |  78.54 |
-- |   6.00 | 113.10 |
-- |   7.00 | 153.94 |
-- |   8.00 | 201.06 |
-- |   9.00 | 254.47 |
-- +--------+--------+
```

-- 🔹 9️⃣ Real-Life Use:
-- ➤ Similar logic can be used for automated table generation in scientific calculations, geometry tools, etc.
-- ----------